

MCMURDO PEGASUS NORTH, Antarctica

WMO: 896670

Lat: 77.950S

Lon: 166.500E

Elev: 10

StdP: 101.20

Time Zone: 12.00 (NZL)

Period: 90-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-47.4	-45.3	N/A	N/A	N/A	N/A	N/A	N/A	24.1	-18.6	21.3	-19.5	1.9	110	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.8	2.4	N/A	1.1	N/A	-0.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	15.0	11.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-21.0	-4.9	-11.9	-21.5	-28.1	-28.0	-28.6	-30.2	-31.3	-29.3	-22.0	-10.8	-4.4		
	DBStd	11.73	3.43	4.25	5.98	7.45	8.00	7.92	7.52	8.29	8.15	5.34	5.31	3.06		
	HDD10.0	11298	462	614	977	1143	1179	1158	1247	1280	1178	991	623	447		
	HDD18.3	14340	720	847	1236	1393	1437	1408	1505	1538	1428	1249	873	705		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.8	3.4	3.7	4.1	4.1	4.3	3.9	3.8	3.5	4.1	3.5	4.4	3.3		
(15) Precipitation	PrecAvg	176	16	22	14	16	22	26	15	12	11	14	12	9		
	PrecMax	343	72	52	60	40	78	134	68	47	35	55	42	22		
	PrecMin	66	0	0	0	0	0	0	1	0	1	0	0	0		
	PrecStd	71	18	14	13	14	20	31	15	12	8	12	12	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	-0.4	-5.8	-6.0	-6.9	-7.5	-8.6	-7.9	-7.7	-7.3	0.6	4.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	2.6	-2.1	-8.0	-11.7	-10.1	-10.1	-11.7	-11.0	-10.2	-9.0	-1.6	2.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	1.4	-4.0	-10.2	-14.7	-13.1	-13.5	-14.8	-14.4	-13.1	-11.8	-2.9	1.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	0.1	-5.6	-12.1	-17.2	-16.5	-16.8	-18.4	-18.1	-16.5	-14.1	-4.0	0.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.6	6.9	7.2	8.8	9.2	9.7	9.5	10.2	9.5	8.7	7.9	6.8		
	5% DB	MCDBR	8.8	8.3	7.5	9.6	10.2	10.8	11.3	11.1	9.6	10.2	8.1	8.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.216	0.173	0.113	N/A	N/A	N/A	0.110	0.164	0.196	0.216	0.233		
	taud		2.228	2.240	2.164	1.892	N/A	N/A	N/A	1.874	2.116	2.222	2.240	2.227		
	Ebn at Noon		964	897	764	103	0	0	0	87	759	927	988	989		
	Edh at Noon		99	77	46	3	0	0	0	3	46	78	98	106		
(44) All-Sky Solar Radiation	RadAvg		8.04	4.64	1.53	0.13	0.00	0.00	0.00	0.02	0.85	3.55	7.38	9.35		
	RadStd		0.22	0.18	0.07	0.01	0.00	0.00	0.00	0.00	0.05	0.13	0.22	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page